

DEEP LEARNING

CORE CONCEPTS HANDBOOK

Artificial Intelligence (AI)

Artificial Intelligence is a branch of computer science that focuses on creating machines or software capable of performing tasks that normally require human intelligence.

These tasks include:

- Understanding language
- Recognizing images
- Solving problems
- Making decisions
- Learning from experience

AI systems try to mimic human thinking and decision-making abilities.

Examples:

- ChatGPT answering questions
- Face unlock in smartphones
- Self-driving cars
- Voice assistants like Siri / Alexa

Machine Learning (ML)

Machine Learning is a subset of AI where computers learn patterns from data instead of being explicitly programmed.

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Instead of writing all rules manually, we provide examples, and the system learns automatically.

Example:

If we give many emails marked spam and non-spam, the model learns how to detect spam emails.

Types:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Deep Learning

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with many layers.

These multiple layers help the system learn highly complex patterns.

It is very powerful for:

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- Images
- Videos
- Speech
- Text
- Medical diagnosis

Example:

A deep learning model can detect pneumonia from chest X-rays.

AI vs ML vs DL

Concept	Description
AI	Broad field of smart machines
ML	Machines learn from data
DL	Uses neural networks with many layers

Relationship:

AI contains ML
ML contains DL

Flow:

AI → ML → DL → Transformers → Large Language Model

Use of Deep Learning :

Traditional Machine Learning works well on small structured data, but struggles with:

- Large data
- Images
- Audio
- Video
- Natural Language

Deep Learning automatically extracts important features.

Example:

Instead of manually telling model “cat has ears, eyes, tail,” deep learning learns these itself.

Biological Neuron

Deep Learning is inspired by the human brain.

A brain neuron has:

- Dendrites → Receive signals
- Cell Body → Process information
- Axon → Sends output signal

Millions of neurons together create intelligence.

Artificial neural networks copy this concept mathematically.

Artificial Neuron

Artificial neuron is the smallest unit of a neural network.

It receives inputs, multiplies them by weights, adds bias, then passes result through activation function.

$$y=f(w_1 x_1 +w_2 x_2 +b)$$

Meaning:

- x = input values
- w = importance of each input
- b = bias
- f = activation function
- y = output

Perceptron

Perceptron is the earliest neural network model used for binary classification.

It gives only two outputs such as:

- Yes / No
- Pass / Fail
- Spam / Not Spam

It is the foundation of modern neural networks.

Inputs, Weights, Bias

Inputs

Data given to model.

Example:

- Study Hours
- Attendance
- Marks

Weights

Weights show importance of each input.

If study hours are very important, weight will be higher.

Bias

Bias is an extra adjustable value helping the model shift output properly.

Activation Function

- Sent

7

Drafts

22

Purchases

22

More
- Labels

+

- Inbox

1,165

Starred

Snoozed

Sent

7

Drafts

22

Purchases

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Activation function decides whether neuron should activate or not.

Without activation, multiple layers behave like simple linear math.

Activation introduces non-linearity, helping model solve complex problems.

Types of Activation Functions

1. Sigmoid

$\sigma(x)=1+e^{-x}1$

Output range: 0 to 1

Used in binary classification.

2. ReLU

$ReLU(x)=\max(0,x)$

If input negative $\rightarrow 0$

If positive $\rightarrow$ same value

Most popular hidden layer activation.

3. Tanh

Range: -1 to 1

Better centered than sigmoid.

4. Softmax

Used for multiclass classification.

Example:

Cat / Dog / Horse

Converts outputs into probabilities.

ANN Architecture

Artificial Neural Network contains:

Input Layer

Receives data.

Hidden Layer

Learns patterns.

Output Layer

Final prediction.

More

Labels

+

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+

Example:

2 Inputs → Hidden Layer → 1 Output

Forward Propagation

In forward propagation, data moves from left to right:

Input Layer → Hidden Layer → Output Layer

Each layer performs calculations and sends result forward.

This generates prediction.

Loss Function

Loss function measures how wrong prediction is.

If actual = 10

Predicted = 8

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Then error exists.

Common Loss Functions:

- MSE
- Binary Cross Entropy
- Categorical Cross Entropy

Lower loss means better model.

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Backpropagation

Backpropagation is the learning mechanism of neural networks.

After prediction:

- Calculate error
- Send error backward
- Update weights
- Improve next prediction

This repeats many times.

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Gradient Descent

Gradient Descent helps reduce loss step-by-step.

It changes weights in correct direction.

Small learning rate = slow learning

Large learning rate = unstable learning

Epoch / Batch / Iteration

Epoch

One complete pass through full dataset.

Batch

Small group of data.

Iteration

One update step using one batch.

Example:

1000 rows, batch size = 100

1 epoch = 10 iterations

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Training vs Testing

Training

Model is fed with training data

-  Inbox 1,165
-  Starred
-  Snoozed
-  Sent
-  Drafts 7
-  Purchases 22
-  More
- Labels +

Testing

Model is checked on unseen data.
Testing tells whether model learned genuinely or only memorized.

Overfitting / Underfitting

Overfitting

Model performs excellent on training but poor on testing.
Reason:
Memorized data.

Underfitting

Model poor on both training and testing.
Reason:
Too simple model.

